Software Requirement Specifications AI-Based Plagiarism Detection Framework

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Version 1.0

1. Executive Summary

This project proposes an AI-powered academic integrity framework that detects direct plagiarism, paraphrased plagiarism, cross-document similarity, and estimates the likelihood of AI-generated text. The system is designed as a decision-support tool for educators rather than a proof engine.

2. Problem Statement

Existing plagiarism tools primarily detect copied text, while AI-generated assignments and advanced paraphrasing make academic integrity harder to assess. The framework combines textual similarity, semantic similarity, and stylometric AI analysis in one platform.

3. Project Objectives

- Detect direct plagiarism using textual similarity.
- Detect paraphrased plagiarism using semantic similarity.
- Identify partial plagiarism at sentence/paragraph level.
- Compare submissions across institutional documents.
- Estimate AI-generated text likelihood using linguistic features.
- Generate explainable reports for faculty.

4. Proposed System Modules

Module	Purpose
Document Processing	Extract text from TXT, PDF, DOCX and preprocess it.
Plagiarism Engine	TF-IDF and semantic similarity analysis.
AI Text Analysis	Stylometric and machine learning based AI likelihood estimation.
Evidence Engine	Highlight suspicious passages with confidence scores.

Reporting Module	Generate explainable reports and recommendations.
Administration Module	Manage datasets, users, and institutional repository.

5. Functional Requirements

1. Upload or paste academic documents.
2. Extract and preprocess text.
3. Detect copied content.
4. Detect paraphrased content.
5. Compare multiple documents.
6. Estimate AI-generated text likelihood.
7. Highlight suspicious passages.
8. Generate downloadable reports.

6. Non-Functional Requirements

- Accuracy through multiple evidence sources.
- Response time under practical academic document sizes.
- Secure handling of uploaded documents.
- Scalable architecture for institutional deployment.
- Explainable output for human review.

7. Technology Stack

Layer	**Technology**
Frontend	React / HTML-CSS-JS
Backend	Python Flask or FastAPI
NLP	spaCy, NLTK, Sentence Transformers
ML	scikit-learn, Hugging Face Transformers
Database	PostgreSQL / MySQL
Storage	Docker

8. Research Components

- Natural Language Processing
- Semantic Similarity
- Stylometry
- Transformer Models
- Explainable AI
- Evaluation Metrics (Precision, Recall, F1, ROC-AUC)

9. Risks & Limitations

- AI detection is probabilistic.
- False positives and false negatives are possible.
- Paraphrased AI text is harder to detect.
- Similarity alone does not prove plagiarism.
- Requires periodic retraining for new AI models.

10. Project Milestones

Milestone's	**Phase/Work**	**Date**
Milestone 01	S.R.S. and Literature Review	21 August 2026
Milestone 02	Version 0.1 (Text/Document Preprocessing)	04 September 2026
Milestone 03	Version 0.2 (Plagiarism Engine)	18 September 2026
Milestone 04	Version 0.3 (Ai Text Module)	02 October 2026
Milestone 05	Version 0.4 (Evidence & Reporting)	16 October 2026
Milestone 06	Integration and Project Review	30 October 2026

11. Deliverables

- SRS document.
- System architecture.
- Prototype web application.
- Trained AI detection model.
- Evaluation report.
- Presentation slides.
- Final technical report.

12. Success Criteria

The project is successful if it detects multiple forms of plagiarism, produces explainable reports, and evaluates AI-generated text using measurable metrics on benchmark datasets.